

A Comprehensive Survey on Automatic Detection of Fake News Using Natural Language Processing: Challenges and Limitations

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Abstract— The study examines how Natural Language Processing (NLP) can be used to automatically detect fake news and how it can be applied to fact-checking in the disciplines of linguistics, computer science, journalism, and information sciences. By evaluating the efficacy, dependability, and breadth of various NLP algorithms, this study demonstrates the possibilities and limitations of autonomous fake news identification. The importance of striking a balance between the technical form of information and its social-cognitive dimensions was revealed by the study. An overemphasis on the technical components can lead to fragmented comprehension, so it is essential to strike a balance between the technical form of information and its social-cognitive dimensions. In the age of digital self-publishing, the importance of authoritativeness in determining the credibility of information has also been raised as a significant concern. The report emphasizes the need for an integrative approach to prevent the spread of fake news, recommending interdisciplinary collaboration and the ongoing refinement of NLP research methods for future studies.

Keywords— Text Analysis; Deep Learning; NLP; Fake News

I. INTRODUCTION

Natural Language Processing (NLP) is a branch of artificial intelligence that focuses on the interaction between human language and computers [1]. NLP techniques apply many feature extraction algorithms to automatically analyze and interpret text data and make it a promising approach for fake news detection [2]. The field of automatic fake news detection has experienced significant growth, leveraging NLP techniques to identify patterns, linguistic features, and other indicators of fake news. By harnessing NLP, researchers aim to develop computational models that can effectively discern false or misleading information within textual content. The growing field of research in fake news detection leverages the power of NLP to automate the identification of false information [3].

This development helps in the fight against misleading information and adds to the preservation of the trustworthiness of information sources. Because of the prominence of social media and online news platforms, substantial alarms have been raised regarding the quick distribution of fake news, which affects individuals, organizations, and governments all over the world. Fake news can have enormous implications, including the spread of misinformation and the potential for significant disruptions to the social and political stability of an area [4]. As a result, it is essential to establish strategies for

identifying and combating fake news in order to maintain a society that is both well-informed and harmonious. In recent years, there has been substantial research dedicated to automating the detection of fake news, leveraging NLP techniques for this purpose [5]. This includes exploring lexical, syntactic, and semantic approaches, as well as employing machine learning and deep learning methods while some studies have shown promising outcomes, several challenges persist in this domain [6]. These challenges encompass the requirement for vast and diverse datasets, the rise of rumors, social bots, and Sybil accounts, and the need to ensure model generalizability across diverse domains and languages such as systems can help individuals, organizations, and governments make more informed decisions, promote media literacy and critical thinking and reduce the harmful effects of misinformation [7]. The goal of this research is to conduct a comprehensive literature review on the topic of applying NLP methods for the automatic detection of fake news. In today's information-based culture, the presence and spread of fake news are becoming an increasing problem. In the worst-case circumstances, widespread panic or harm might result from the mass dissemination of false information [8]. This research makes several important contributions to the fields of fake news identification and NLP, including the following. This study presents a thorough analysis of the current NLP techniques used for the automatic detection of fake news, classifying and discussing each technique in great depth. The merits and disadvantages of each method are thoroughly discussed, as are their respective strengths. Understanding the constraints can help direct future research for development and innovation, while the strengths emphasize the promise and efficacy of NLP in this setting. Comparative Analysis: It provides a comparison of various NLP-based fake news detection algorithms, illuminating their efficacy, accuracy, and dependability. This paper recommends future research directions by pointing out gaps and obstacles in the existing environment of fake news identification. The findings of this study have important policy and practical consequences for internet companies, social media platforms, news organizations, and legislators working to combat the spread of disinformation. By making the poll easily accessible, this study aids in raising public awareness of the dangers of fake news and the methods for detecting it. In today's technologically advanced world, the ability to effectively use digital tools is a fundamental skill [9].

The purpose of this work is to demonstrate how NLP can be applied to the problem of detecting fake news by evaluating text features and patterns that may signal the news's authenticity or lack thereof. Better search engine optimization and content strategy are two areas where NLP has proven useful ;the rest of the paper is structured as follows: in Section 2, we will review some of the works done in fake news detection and provide a summary of these works, Section 3 will contain the applications of NLP and how fake news can be detected using NLP applications, in Section 4 we will provide a survey of challenges and limitations of NLP applications in fake news detection and finally, we will conclude this paper in section 5 and open the door for future research on this matter.

II. LITERATURE

Detecting fake news automatically presents numerous challenges that extend beyond traditional content-based analysis methods. One key hurdle lies in the fact that comprehending news articles often entails understanding the underlying political or social context, as well as relying on 'common sense' knowledge elements that current NLP algorithms struggle to capture effectively. However, recent research indicates that fake and genuine news exhibit distinct dissemination patterns on social media platforms, which can be leveraged to develop automated systems for identifying fake news. This literature review explores the latest advancements in fake news detection, focusing on the utilization of propagation patterns and social media dynamics to enhance the accuracy and efficiency of automated fake news detection algorithms.

Reference [7], introduce a novel approach for automatically detecting fake news using geometric deep learning. The model extends traditional Convolutional Neural Network (CNN) to handle graph data, allowing us to integrate various types of information like content, user profiles, social connections, and news propagation. They trained and tested our model on a dataset of verified news stories shared on Twitter. The results indicate that the structure of social networks and the way news spreads are crucial for achieving highly accurate 92.7% Receiver Operating Characteristic (ROC) fake news detection. Furthermore, their findings suggest that fake news can be reliably detected within a few hours of its propagation.

In [8], Kesarwani et al. introduce a straightforward method for detecting fake news on social media utilizing the K-Nearest Neighbor (KNN) classifier. The approach achieved a classification accuracy of approximately 79% when tested on a dataset of news posts from Facebook. By leveraging the KNN classifier, they demonstrate the potential to effectively identify fake news in social media contexts, contributing to ongoing efforts in combating misinformation.

In [9], Mansouri et al. proposed a semi-supervised learning framework that leverages both labeled and unlabeled data, employing a CNN. The method involves extracting various features from text and image data using the CNN, followed by the utilization of Linear Discrimination Analysis (LDA) to classify unclassified data. Additionally, the fitness function is adapted to enhance the impact of estimated classes at each step. The results demonstrate that the proposed method

surpasses other approaches in terms of recall, specificity, and sensitivity, achieving an impressive precision value of 95.5%. This highlights the effectiveness of utilizing deep learning techniques for accurate fake news detection.

Reference [10], used Bidirectional Encoder Representations from Transformers (BERT) to produce a new model that can detect fake news based on transfer learning to detect bot accounts in a COVID-19 dataset and then make improvements to the model. The dataset they depend on is the CoAID dataset. They collected the tweets by the IDs of users when reaching the attributes of the users. Also, they did not use LDA to filter the tweets. Furthermore, the researchers incorporated the Cresci 2017 dataset to fine-tune the BERT model. By leveraging this dataset, they trained the BERT model to classify COVID-19 tweets into two categories: bot and not bot. Using the new model for fake news detection, they achieved an accuracy of 80%.

Authors in reference[11], Jain et al. proposed an approach to detect fake news in web-based network media by using a new strategy depending on the use of Naïve Bayes classifier, Support Vector Machine, and semantic investigation. With a hybrid approach, they handle recognizing the category of news, which is not easy, and they provide better results than other studies by achieving 93.50% accuracy.

Proposed a stacking mechanism consisting of three deep learning combined approaches based on detecting fake news. The connected models are CNN, Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU). Also, besides deep learning, they used machine learning models like Support Vector Machines (SVM) and Decision Trees (DTs) to be a powerful model achieving an accuracy of 99.94% and 96.05%, respectively, on the dataset and KDnugget dataset [12].

TABLE I. SUMMARY OF LITERATURE

Ref	MSE	Accuracy (%)	Implementation Method
[7]	0.19	92.7	Geometric deep learning, extension of CNNs to handle graph data
[8]	0.43	79%	K-Nearest Neighbor (KNN) classifier
[9]	0.03	95.5	Semi-supervised learning framework using CNN and LDA
[10]	0.31	80	Transfer learning with BERT, using CoAID and Cresci 2017 datasets
[11]	0.12	93.50	Hybrid approach with Naïve Bayes classifier, Support Vector Machine, and semantic analysis
[12]	0.02	99.94	Stacking mechanism combining CNN, LSTM, GRU, and machine learning models
[13]	0.72	70.6	FDML model with multi-task learning and N-Graph for fake news detection

In [13], Liao et al. introduced a novel multi-task learning model for the detection of fake news using the LIAR dataset based on two types of fake news. The first type is some topics have higher percentages of fake news, and the second one is some news authors have higher intentions to publish fake

news. The Fourier Domain Mode Locking (FDML) model can provide topic labels for fake news and provide contextual information for the news to increase the performance of short fake news detection. The model has been designed based on two parts: In addition, they introduced a new proposal for the News Graph (N-Graph) to preserve the relationships in which news is linked by relying on text information and contextual information simultaneously. They compared their proposed model with other models to find the accuracy achieved is 0.706%, while other models like Random Forest (RF) and LSTM are in the range of 60-65% accuracy. Table 1 shows the summary of the literature view.

III. NLP APPLICATIONS

The study of how computers may learn to read, comprehend, produce, and generally function with human language is the focus of NLP, a sub-field of artificial intelligence (AI). It enhances the capability of computers to comprehend human language and vice versa. The field of NLP combines statistical and deep learning models with those of machine learning and computational linguistics, which models human language based on rules. The application of natural language processing can be observed in various areas: Applying natural language processing to text data, such as posts on social media or product reviews, enables the analysis of customers' emotional states to be obtained [14].

NLP is crucial to the operation of chatbots on different customer care platforms and virtual assistants (Fig. 1), like Google Assistant, Siri, and Alexa. They use natural language processing to interpret user input and act accordingly. Applications like Google Translate use NLP to translate written or spoken content between languages. To facilitate faster interpretation and evaluation of information, NLP is used to construct brief summaries of lengthy papers. names, locations, dates, and other key information can be isolated from massive amounts of text using natural language processing. This is vital for ensuring legal compliance, conducting market research, and gathering intelligence. Speech recognition technology is heavily dependent on NLP to transcribe human speech into text and interpret its meaning [14]. This has practical uses in areas such as voice-enabled controls and transcription services [15]

Fig. 1 illustrates the branches of NLP as a sub-field of artificial intelligence and computational linguistics that focuses on the interaction between computers and human language. NLP encompasses a wide range of topics and techniques aimed at understanding, analyzing, and generating natural language text. Here are some key topics in NLP:

- **Tokenization:** The process of breaking down text into smaller units, such as words or sub-words, known as tokens. Tokenization is a fundamental step in NLP as it provides the basis for further analysis and processing.
- **Part-of-Speech Tagging:** Assigning grammatical tags to each word in a sentence, such as a noun, verb, adjective, etc. Part-of-speech tagging helps in understanding the syntactic structure of a sentence and is used in various NLP applications like machine translation, sentiment analysis, and information retrieval.
- **Named Entity Recognition (NER):** Identifying and classifying named entities in text, such as person names, organization names, locations, dates, and more. NER

plays a crucial role in tasks like information extraction, question answering, and text summarization.

- **Sentiment Analysis:** Analyzing text to determine the sentiment or emotion expressed, typically categorized as positive, negative, or neutral. Sentiment analysis finds applications in social media monitoring, customer feedback analysis, and brand reputation management.
- **Topic Modeling:** A statistical technique to discover latent topics or themes in a collection of documents. Topic modeling algorithms, such as Latent Dirichlet Allocation (LDA), can uncover the underlying themes in text corpora and assist in tasks like document clustering, content recommendation, and trend analysis
- **Text Classification** Categorizing text into predefined classes or categories. Text classification is widely used in spam filtering, sentiment analysis, news categorization, and document classification tasks.
- **Machine Translation:** The task of automatically translating text from one language to another. Machine translation systems use various techniques, including statistical methods, neural networks, and rule-based approaches to facilitate cross-lingual communication.
- **Question Answering:** Building systems that can understand questions posed in natural language and provide relevant answers. Question answering involves techniques like information retrieval, passage ranking, and natural language understanding to provide accurate responses.
- **Language Generation:** Generating human-like text based on a given prompt or context. This includes tasks like text summarization, dialogue generation, and machine-authored content.

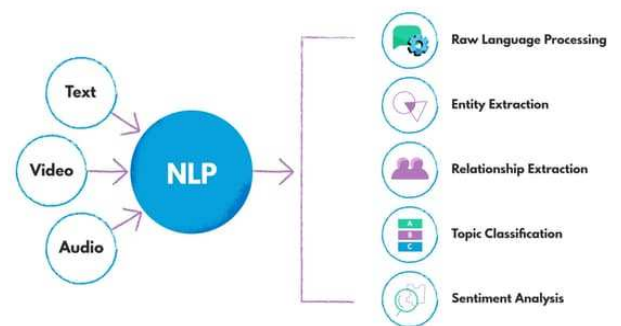


Figure 1. NLP applications [6].

Differentiating between truth and lies, two realms in which linguistics has continued to make contributions, is all that is required to distinguish reliable information from unreliable information [16]. It is, therefore, rather logical that automatic systems for identifying false news borrow from the same field of linguistics its theories and models, the application of which in the context of enormous amounts of data is not so evident. In point of fact, the development of linguistic systems for identifying erroneous information has been aided by the predictive analysis of written texts. 'The interrelation between linguistic markers and action considered as an intentional, conscious, and subjectively significant activity' is what this research focuses on" [17].

To put this into more concrete terms, the predictive analysis model utilized in detection systems begins with the

hypothesis that the majority of liars use strategic verbal or written language to deceive their audience. Even when they make an effort to control their speech, "language leaks" nevertheless occur. These "language leaks" can be recognized by certain verbal features, most notably the frequency of words, the use of pronouns, conjunctions, and phrases that have unpleasant emotions associated with them. The "gaps" detected in the substance of a message are precisely what the linguistic approach is looking for to accomplish its goal.

The application of machine learning technology is seen as the primary focus in this method. Systematized approaches to the detection of fake news outline a variety of procedures, the most prominent examples of which are:

- Text representation: Considered to be the simplest technique in this sense, it is only interested in words taken individually, aggregated, and analyzed to reveal signs of deception. A basic approach that can sometimes be enriched by additional indexing of keywords (thematic index, geographical index), or a shallow syntactic analysis [18].
- In-depth syntactic analysis (Deep Syntax): In Automatic Language Processing (TAL), the third level of analysis is after those known as surface semantics and syntactics. It is not enough to analyze the words superficially but also study the sentences that contain them as well as their syntactic structure. Deep parsing is implemented through the Statistical Model PCFG (Probability Context-Free Grammars) [19].
- Semantic analysis: Proposes to study the meaning, the signified; and for that, it needs support, in this case, the signifier, the form. It extends the performance of syntactic analysis by searching for veracity indices obtained by analyzing the degree of compatibility between the message contained in the text and the profile of its writer. This approach assumes that a credible author's text on a given topic should contain roughly the same consistency and information as conveyed in the texts of other credible writers on the topic. It is therefore important, for the comparison, and therefore for its success, to have a database of similar content previously verified [20].

Discourse analysis (Rhetorical Structure Theory - RST): Proven methods of discourse analysis did not lend themselves to the automatic generation of texts, at least until the 1980s when researchers at the Institute of Information Science of the University of Southern California, noting this gap, have built, from studies of elaborated texts, the Theory of Rhetorical Structure. Nowadays, independently of its use in computer science and the treatment of masses of data, it has acquired a special status in linguistics. The RST, as presented by the latter, offers an explanation of the consistency of the texts, that is to say, 'the absence of illogicality and gaps'; an essential function that fake news detection systems cannot overlook. The application of rhetoric to digital information can be indicative of deception, to be sure, but its application currently remains relatively limited [21]

IV. LIMITATION ANDS CHALLANGES

In view of the preceding developments, it is appropriate to recognize the important place occupied by the verification of information in research programs in linguistics, computer science, journalism and also in information sciences. This can

constitute a difficulty for an objective criticism, which would suppose that one respects the epistemological field of the criticized model [21]. The ideal critical analysis would try to rise above the chapel debates and would try to detect in each model the elements that contribute to an effective search for the truth. Faced with this impossibility, it would not be less relevant to favor the Information and h9Communication Sciences approach, which will thus require us to refer to the information evaluation criteria developed by researchers in this field. To do this, should we first agree on a definition of information [22]. The first information theoreticians, mathematicians, presented it in its technical, transmissible form, that is to say as a signal and not as a symbol, a meaning, a content [23].

Complementing this technical view of information, researchers in the humanities and social sciences have proposed definitions that take into account the social and cognitive dimensions of information. According to [24] information cannot be restricted to its technical form. It can designate data, information, a fact, and a message; it is the essential element on which knowledge is based. It integrates both written processes, oral and visual processes, and is exchanged between individuals, in societies [25].

It is an essential need for life in society. CIS are thus interested both in the information and in the condition of its transmission. At a time when digital information dominates, it is more common to reduce information to its technical meaning. Indeed, the use of Machine Learning technology in linguistic approaches to detecting fake news (as shown in Fig. 2.) is certainly comfortable given the enormous amounts of information to be processed, but suffers from a shortcoming inherited from the Shannonian conception of information. The cognitive and social attributes of information, essential in their understanding, seem to be obscured. The computing power of machines contrasts with their difficulty in perceiving these dimensions Under these conditions, any search for truth seems biased [26].

For example, analyzing the effectiveness of the detection method by text representation, noted an inefficiency linked in particular to the fact that it is based on an extraction of words dissociated from their context; which makes any perception of meaning impossible. This makes the authors say that this technique would gain in efficiency if it were coupled with additional analysis techniques [27].

As shown in Fig. 2. Detecting fake news using NLP involves analyzing textual information to identify patterns, biases, and other indicators that can suggest the presence of misinformation or disinformation. Here's a general approach to detecting fake news using NLP: Collect data for the purpose of creating a reliable dataset: Obtain a dataset with labeled news items or statements, each of which should specify whether it is real or not. This will allow you to evaluate the authenticity of the information. On the internet, you can get datasets of this kind, such as the one that was generated by the Fake News Challenge or the one that was generated by PolitiFact. To get the data ready in advance, remove any extraneous information from the text data, such as HTML elements, punctuation, and stop words, and then verify that the data is in a clean and usable format. The process of tokenization is what needs to be done in order to break the text down into its individual words, also known as tokens. You might alternatively normalize words by converting them back

to their lemmatized or stemmed form by utilizing stemming or lemmatization. This would be another way to normalize words. The separating of features includes: Determining what the important parts of the text are and then pulling them out. A few examples of typical features include word frequency, n-grams (which are continuous sequences of words), TF-IDF (which stands for term frequency-inverse document frequency), and named entities. These characteristics will be useful in putting the textual information in a manner that can be understood by machine learning algorithms, which is numerical. The portrayal will end up being more accurate as a result of this. Train a model by employing machine learning by making use of the data that has been preprocessed and the features that have been retrieved when training a classification model. Naive Bayes, Logistic Regression, Random Forest, and Support Vector Machines are some of the algorithms that come to mind when thinking about the most popularly utilized ones for detecting fake news. In order to get the most accurate predictions from the model, it is strongly suggested that you use a cross-validation technique in conjunction with a feature selection method. This will help you get the most accurate predictions possible. Consider both the lexical and the contextual aspects. Techniques based on natural language processing can potentially make use of linguistic and contextual indicators to identify fake news stories. Take into consideration completing an examination of the material's tone, establishing how readily it can be read, searching for assertions that are too extreme, and determining how reputable the source is. You have the choice of applying rule-based methods or pre-trained models, such as sentiment analysis, when carrying out these kinds of assessments. One of the options is presented below. Verification of the facts and the gathering of information from other sources: Your detection pipeline should consist of both internal methods for fact-checking and external sources for information. Make use of the fact-checking platforms and APIs that are already accessible in order to evaluate the statements made in the news article in light of the findings obtained from reputable sources. Through the process of cross-referencing material and detecting discrepancies, additional evidence of the existence of false news can be gained. Always keep a watchful eye on social media and the content that is offered by users: Your research should include not only news articles but also social media websites, as it is on these platforms that erroneous information often spreads at a fast pace. It is crucial to keep an eye on user-generated content, study patterns of content distribution, and apply technologies such as sentiment analysis and network analysis in order to discover potentially hazardous information. This will allow you to identify information that could be harmful. Conduct an analysis, and if necessary, make changes to the model: Assessing how effectively your model is functioning can be done through the use of assessment measures such as accuracy, precision, recall, and F1 score. You have the option of generating a brand-new validation dataset or making use of methodologies such as cross-validation in order to acquire an accurate evaluation of how well the model functions. Incorporate feedback and check to see that the training data are kept current at all times in order to ensure that the model can undergo ongoing improvement. It is crucial to bear in mind that recognizing fake information is a challenging task, and there is no one approach that can guarantee one hundred

percent precision. This fact should be kept in mind at all times. Combining a wide variety of tactics, methods, and strategies can make the detection process more effective as a whole. This can be accomplished by gathering as much information as possible.

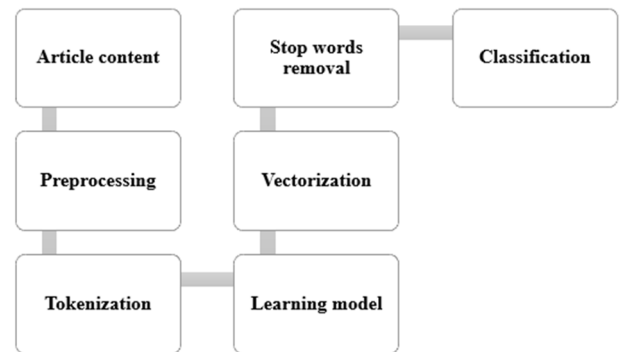


Figure 2. General workflow of ML based detection models [28]

The same critique applies to deep parsing, which, according to the opinion of [28], can be interesting when complementing the text representation method. However, when used alone in the detection of fake news, it is considered insufficient. Indeed, the combination of deep syntactic analysis with other linguistic analyses allows, according to these authors, an efficiency rate of around 90%, provided this method is applied to a single database. This effectiveness, however, should be viewed in light of the author's preference for a technical approach to information. Therefore, the ongoing epistemological criticism boils down to a fundamental question: Is it enough to assign reliability percentages to words or phrases to declare information reliable [29].

On the other hand, semantic analysis devices introduce a measure of credibility to the authors of messages by comparing the profiles of authors of similar messages. This deviates from the trend observed in previous technical approaches and represents a form of socialization of verification linked to documentary techniques of information validation. In this context, the credibility of the author becomes an indicator of the quality of information sources, alongside criteria such as relevance, exhaustiveness, precision, completeness, and reliability [30].

However, with the "reversal of the editorial chain," the status of an author is no longer conferred by an obligation to go through the classic circuit of publishing and marketing a work. Self-publishing, where consumers have positioned themselves as producers of information, has transformed every web user into an author. From this perspective, the notion of an author, considered "the first cause of everything," is only relevant when it designates the status that makes an author an author and what establishes its value, making an author validated or not [31].

The existence of a filtering system gives credit to the author and allows them to be an authority in the field. Outside this validation circuit, as seen in certain social publication platforms, it is more appropriate to speak of "authoritativeness": the attitude of producing and making public texts, self-publishing, or publishing on the WWW

without going through the consent of reference institutions tied to the printed order [32].

Authoritativeness allows for self-proclaimed authorship but does not inherently confer any value on either the publication or its author. Therefore, considering the comparison of editor profiles implemented in semantic devices for detecting fake news as a sufficient criterion for the reliability of information is challenging. Additionally, the use of this method cannot be generalized, as its application must be restricted to a single domain. Its effectiveness depends on the availability, timeliness, and completeness of data and the ability to describe and classify the authors associated with the content. The limitations and key constraints of the detection methods can be summarized in Table 2 below [33]:

TABLE II. THE KEY LIMITATION EACH CHALLENGE TYPE STYLES

Challenge	Limitation
Interdisciplinary nature of information	Verification of information is important across linguistics, computer science, journalism, and information sciences, due to the interdisciplinary nature of information.
Technical vs Social Cognitive View of Information	Initial definitions of information were technical, focused on transmissible form. However, researchers in humanities and social sciences have highlighted the importance of the social and cognitive dimension of information.
Issues with Machine Learning	The use of machine learning in detecting fake news can be efficient due to the large volume of data, but it lacks an understanding the cognitive and social attributes of information.
Text Representation Method	The method based on text representation is efficient when combined with other techniques but insufficient when used alone, as it does not consider context.
Epistemological Criticism	Questions the reliability of information based solely on word or phrase reliability percentages.
Semantic Analysis	Introduces credibility measures for message authors by comparing profiles. The credibility of the author serves as an indicator of information source quality.
Author vs. Authoritativeness	In the digital age, the concept of 'authoritativeness' emerges as anyone can produce and publish text without traditional validation, thereby questioning the reliability of information.
Limitations of Semantic Devices	Semantic devices for detecting fake news may not be sufficient for information reliability. Their effectiveness depends on the availability, timeliness, and completeness of data, and the ability to describe and classify associated authors.

V. CONCLUSIONS

In this research delves deeper into the utilization of Natural Language Processing (NLP) technologies for the automatic detection of fake news. Given its pivotal role in various academic fields, including linguistics, computer science, journalism, and information sciences, the verification of information emerges as an intriguing and crucial topic. While NLP holds promise for the future in this domain, our investigation has uncovered both its strengths and areas requiring further research. Striking a balance between the technological form of information and its social-cognitive component is crucial, as NLP-based algorithms show

significant potential but also have fundamental limitations. Maintaining this equilibrium is vital for progress in addressing the challenging issue of fake news.

Our research findings suggest that techniques such as semantic analysis, deep parsing, and text representation have shown promise in the past, but they are not without flaws. Overemphasis on the technical form of information at the expense of its social and cognitive dimensions poses obstacles to achieving comprehensive comprehension and detection. Additionally, concerns persist regarding the veracity of data based solely on reliability percentages for specific words or phrases. Therefore, there is a need to critically assess these approaches and continuously enhance them.

We emphasize the importance of 'authoritativeness' in the digital age as a new dynamic that must be considered when evaluating the believability of information. The traditional conception of authorship has evolved with the rise of self-publishing, adding complexity to the challenge of identifying fake news. In light of these findings, further research is warranted to investigate these challenges and refine the application of natural language processing (NLP) in fake news detection. We advocate for expanded collaboration across academic fields, particularly in the technical and social sciences, to better understand information as a whole and address the issue of fake news.

The significance of this area of investigation is underscored by our findings, considering the social, political, and economic effects caused by fake news in today's society.

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